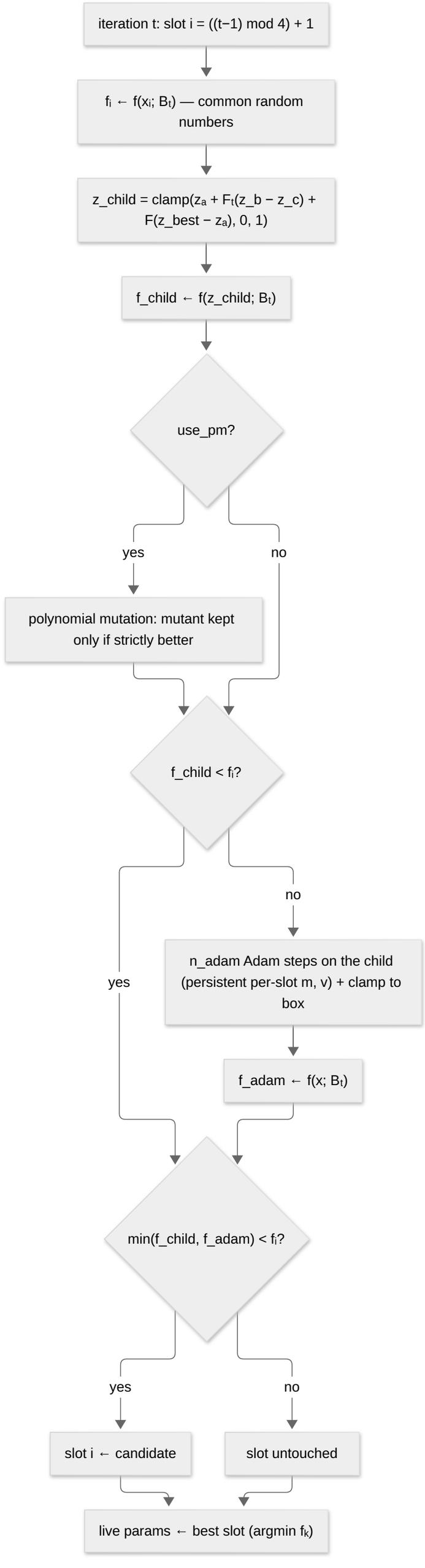

iteration t: slot i = ((t−1) mod 4) + 1
$f_i \leftarrow f(x_i; B_t)$ — common random numbers
$z_child = clamp(z_a + F_t(z_b - z_c) + F(z_best - z_a), 0, 1)$
$f_child \leftarrow f(z_child; B_t)$
use_pm?
yes
no
polynomial mutation: mutant kept only if strictly better
f_child < f_i?
yes
no
n_adam Adam steps on the child (persistent per-slot m, v) + clamp to box
$f_adam \leftarrow f(x; B_t)$
min(f_child, f_adam) < f_i?
yes
no
slot i ← candidate
slot untouched
live params ← best slot (argmin f_k)